

Local Scale Adaptation for Augmenting Hand Shape Models

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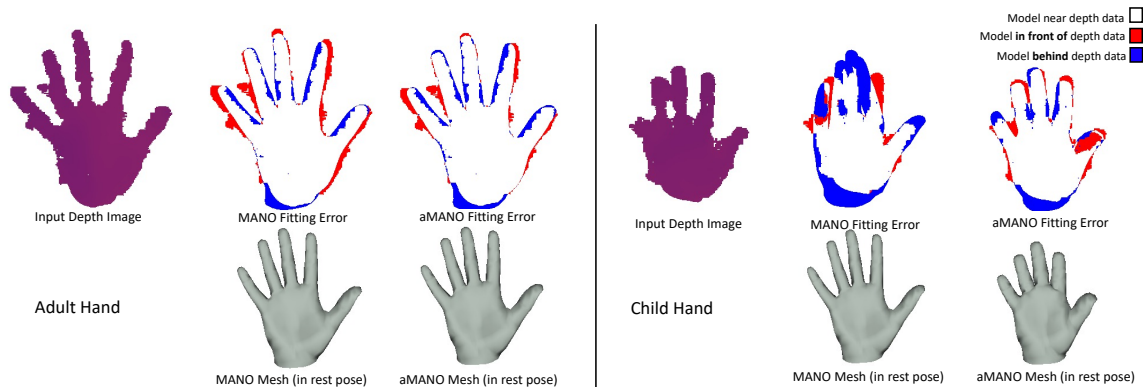


Figure 1: Existing hand shape models, e.g. MANO, fit well to adult hands (left column), but do not fit well on child hands (right column). Our proposed hand shape model aMANO (right) fits to both adult and child hands compared to MANO (center). The bottom row shows the calibrated rest pose MANO mesh (left) and aMANO mesh (right).

ABSTRACT

The accuracy of hand pose and shape recovery algorithms depends on how closely the geometric hand model resembles the user's hand. Most existing methods rely on learned shape space, e.g. MANO; but this shape model fails to generalize to unseen hand shapes with large deviations from the training set. We introduce a new hand shape model, aMANO, that augments MANO by introducing local scale adaptation that enables modeling substantially different hand sizes. We use both MANO and aMANO for calibrating the shape to new users from a stream of depth images and observe the improvement of aMANO over MANO. We believe that our new hand shape model is a significant step in improving the robustness and accuracy of existing hand tracking solutions.

CCS CONCEPTS

• **Computing methodologies** → **Reconstruction; Mesh models.**

KEYWORDS

hand shape, hand pose, hand tracking, local scale adaptation

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1 INTRODUCTION

Hand pose and shape estimation is an essential component for interaction in many applications. Human hands occur in a variety of shapes and sizes, and admit a large pose configuration space, which makes estimating the shape and pose of the user's hand from single or even multiple views, ill-posed. It has been well established in the literature [Tan et al. 2016] that a user-specific (calibrated) hand model is essential for accurate hand shape recovery.

Our underlying hand model is a mesh-based model, MANO [Romero et al. 2017], which is the most widely used hand model in literature. However, this data-driven hand model cannot adapt gracefully to unseen hand shapes with large deviations from the training set, e.g. a child's hand, which adversely affects tracking (see Fig. 1 right column).

We propose *adaptive MANO (aMANO)* that extends the shape space of MANO by introducing local scale adaptation. The simplicity of aMANO makes it easy to integrate into existing hand shape and pose recovery frameworks. Specifically, we add both aMANO and MANO into the same existing framework in order to establish better hand pose and shape recovery accuracy, on a wide variety of challenging datasets.

Contribution. The primary contribution of this paper is to present aMANO, a novel hand shape model obtained by adapting the existing MANO model using local scale parameters that allows it to fit to unseen hand shapes with large deviations from the original MANO shape space.

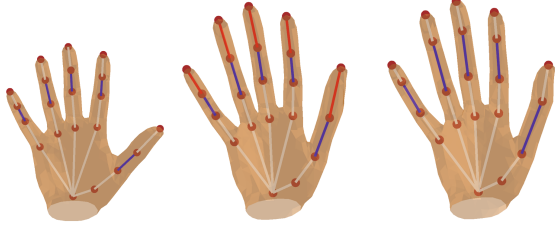


Figure 2: We double the scale for the bone connecting the PIP joint to DIP joint (marked in blue) of the original mesh (left). We compare the deformation in directly scaled LBS (middle) with LBS modified with endpoint weights (right). Notice the unwanted scaling of the bone connecting the DIP joint to the TIP (marked in red in middle image) is avoided using the modified LBS.

2 HAND MODEL

The PCA shape space of the original MANO model is learned from 2018 scans of 31 subjects. As a result it cannot fit to unseen hand shapes with large deviations from training data. We augment the shape space of MANO by introducing local scale parameters ϕ that compliments the original shape parameters β and thus, increases the span of the shape space.

Specifically, our local scale parameters are based on the idea of stretchable bones introduced by [Jacobson and Sorkine 2011]. We assume a set of local scale parameters $\phi \in \mathbb{R}^{n_b}$ for each of the n_b bones in the hand. The scale factor for each bone ϕ_j is calculated as

$$\phi_j = \frac{\|a'_j - b'_j\|}{\|a_j - b_j\|} \quad (1)$$

where $a_j \in \mathbb{R}^3$ and $b_j \in \mathbb{R}^3$ are the start and end positions of the original bone and a'_j and b'_j are the start and end positions of the scaled bone in the posed mesh. The bone endpoints are the keypoints in the hand model (see Figure 2).

We use a modified linear blend skinning (LBS) to incorporate the local scale parameters ϕ when deforming the mesh vertices v using the pose parameters θ . Using the modified LBS, each deformed vertex v'_i is given by

$$v'_i = \sum_{j=1}^{n_b} W_{b_{ij}} \left\{ a'_j + R_j \left(W_{e_{ij}} s_j + (-a_j + v_i) \right) \right\} \quad (2)$$

where $s_j = (\phi_j - 1)(b_j - a_j)$, $R_j \in \mathbb{R}^{3 \times 3}$ is the rotation matrix that takes bone j 's rest vector $(b_j - a_j)$ to its pose vector $(b'_j - a'_j)$, v_i is i^{th} vertex of the rest pose mesh with shape parameter β and $W_b \in \mathbb{R}^{n_v \times n_b}$ are the bone weights. An additional set of endpoint weights $W_e \in \mathbb{R}^{n_v \times n_b}$ for each bone is required to restrict the effect of scaling to part of the mesh associated with the scaled bone. This is illustrated in Figure 2. We compute these endpoint weights using bounded biharmonic weights [Jacobson et al. 2011]. Here $R_j(\theta)$ is computed around fixed axes of rotation by recursively accumulating along the kinematic hierarchy with pose parameters θ .

Table 1: $d2m_{max}$ comparison for MANO and aMANO on examples from various standard datasets and our captured dataset.

User	MANO $d2m_{max}$ (mm)	aMANO $d2m_{max}$ (mm)
NYU [Tompson et al. 2014], User 1	23.2	18.4
NYU [Tompson et al. 2014], User 2	23.6	19.1
BigHand [Yuan et al. 2017], User 1	27.7	21.6
BigHand [Yuan et al. 2017], User 2	23.8	16.0
GuessWho [Tkach et al. 2017], User 1	19.5	14.2
Real User 1	17.8	13.6
Real User 2 (Child)	9.2	7.2

3 EVALUATION

We quantitatively evaluate our shape and pose fitting using the dense data-to-model metric $d2m$. To capture robustness, we compute the maximum value of the metric over points in the observed point cloud, and average it over all frames of a user to get the value for $d2m_{max}$

$$d2m_{max} = \max_{p \in D(data)} \|p - \Pi_{D(model)}(p)\| \quad (3)$$

We present the shape and pose fitting accuracy for the original MANO and our proposed aMANO model on examples from NYU [Tompson et al. 2014], Bighand2.2M [Yuan et al. 2017], Guess-Who [Tkach et al. 2017] and our captured dataset in Table 1. We observe that our aMANO model fits significantly more accurately than the original MANO model on all these datasets. Lower values for $d2m_{max}$ are better.

4 CONCLUSION

Accurate hand shape and pose fitting rely on a well-calibrated hand model. We propose a new hand model that augments the shape space of the existing hand model to fit to a larger demographic. Our hand model is easy to integrate with existing hand shape and pose recovery methods and outperforms the most widely used hand model in current literature. Our current local scale parameters can only adapt along the bones, however, we can further increase the shape space by introducing the ability to vary the thickness of fingers and palm.

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